Modeling, Segmentation & Evaluation

Overview

This notebook focuses on building, comparing, and evaluating machine learning models for the Bank Marketing dataset. Building on the preprocessing and feature engineering steps from previous notebooks, we now develop predictive models to estimate the likelihood of a client subscribing to a term deposit.

In addition to a traditional global modeling approach, this notebook explores a segmentation-based strategy using clusters derived earlier. By comparing a single global model against cluster-specific models, we aim to understand whether segment-level specialization leads to improved predictive performance and better business outcomes.

Objectives

- Compare multiple candidate model architectures using the training and validation sets
- Select a primary modeling approach based on validation performance
- Train a global model using all available data and evaluate performance across clusters
- Train segment-specific models for each cluster and compare against the global model
- Perform hyperparameter tuning for both global and segment-specific models using validation data
- Train final models using selected hyperparameters
- Apply probability calibration to improve predicted probabilities

Dataset Description

The modeling process uses the preprocessed dataset generated in the previous notebook, which includes engineered features such as categorical encodings, binary indicators, and interaction terms. Additionally, each observation is assigned to a cluster derived from unsupervised learning, enabling both global and segment-specific modeling approaches.

The data is split into three subsets:

- **Training set**: used to fit models
- **Validation set**: used for model selection, hyperparameter tuning, and calibration
- **Test set**: held out for final unbiased evaluation

The target variable represents whether a client subscribed to a term deposit.

Key Considerations

- **Model comparison**: Different model architectures are evaluated to identify the most suitable approach for the problem
- **Segmentation strategy**: Cluster-based segmentation allows us to investigate heterogeneity in client behavior and model performance
- **Fairness and consistency**: Performance is evaluated across clusters to detect potential disparities
- **Overfitting prevention**: Validation data is strictly used for model selection and hyperparameter tuning
- **Probability calibration**: Raw model probabilities may be poorly calibrated, which can negatively impact decision-making

Outcome

By the end of this notebook, we will have: - Selected and trained a primary global model
- Developed segment-specific models tailored to different client clusters
- Tuned and calibrated all models for improved predictive reliability
- Compared global and segmented approaches from both predictive and business perspectives

- Identified optimal decision thresholds based on expected cost

```
import pandas as pd
import lightgbm as lgb
import optuna
import joblib
import seaborn as sns
import matplotlib.pyplot as plt

from pathlib import Path
from sklearn.model_selection import ParameterGrid
from sklearn.isotonic import IsotonicRegression
from sklearn.metrics import (
    average_precision_score,
    precision_score,
    recall_score,
    f1_score,
)
import numpy as np
import pandas as pd


from utils.utils import (
    load_dataset,
    evaluate_model,
    evaluate_from_probs,
    plot_cluster_pr_curves,
)

RANDOM_STATE = 705
```

```
/Users/tonantzinrealrojas/miniforge3/envs/nlp/lib/python3.13/site-packages/tqdm/auto.py:21: T
  from .autonotebook import tqdm as notebook_tqdm
```

```
# quarto preview 03_training.ipynb --to pdf
# quarto render 03_training.ipynb
# black 03_training.ipynb
```

1. Global architecture selection

In this section, we evaluate a set of candidate model architectures to determine which will be used in the subsequent comparison between a global model (trained on all observations) and segment-specific models (trained per cluster).

We begin by training and assessing five baseline architectures using their default configurations:

1. Logistic Regression
2. Random Forest
3. XGBoost
4. LightGBM
5. CatBoost

The goal is to identify a strong primary model that balances predictive performance and generalization before proceeding to more advanced analysis.

```
import numpy as np
import pandas as pd

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# from sklearn.metrics import (
#     average_precision_score,
#     f1_score,
#     precision_score,
#     recall_score,
# )

from xgboost import XGBClassifier
import lightgbm as lgb
from catboost import CatBoostClassifier
```

Because the outcome variable is highly imbalanced, we apply class weighting across all models to penalize misclassification of subscribers more heavily than non-subscribers. For tree-based models, this is implemented via the ratio of negative to positive samples, while linear and ensemble baselines use built-in balanced class weights. This approach preserves the original

data distribution and produces probability estimates suitable for downstream calibration and cost-based decision analysis.

```
X_train_, y_train = load_dataset("02", "train")
X_validation_, y_validation = load_dataset("02", "validation")
```

```
train
X shape: (28934, 60)
y shape: (28934,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	0.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0

```
validation
X shape: (7234, 60)
y shape: (7234,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

```
cluster_feat = "cluster"

X_train = X_train_.copy()
X_train = X_train.drop(columns=cluster_feat)
print("\nTrain set: ", X_train.shape)
display(X_train.head(3))

X_validation = X_validation_.copy()
```

```
X_validation = X_validation.drop(columns=cluster_feat).copy()
print("\nValidation set: ", X_validation.shape)
display(X_validation.head(3))
```

```
Train set:  (28934, 59)
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__job_
0	0.0	0.0	0.0	0.0	0.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0

```
Validation set:  (7234, 59)
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

```
n_pos = y_train.sum()
n_neg = len(y_train) - n_pos
scale_pos_weight = n_neg / n_pos
```

i. Logistic Regression

```
log_reg = LogisticRegression(
    penalty="l2",
    solver="liblinear",
    class_weight="balanced",
    max_iter=1000,
    random_state=RANDOM_STATE,
)

log_reg.fit(X_train, y_train)
log_reg_metrics = evaluate_model(log_reg, X_validation, y_validation)
```

ii. Random Forest

```
rf = RandomForestClassifier(
    n_estimators=100, class_weight="balanced", random_state=RANDOM_STATE, n_jobs=-1
)

rf.fit(X_train, y_train)
rf_metrics = evaluate_model(rf, X_validation, y_validation)
```

iii. XGBoost

```
xgb = XGBClassifier(
    objective="binary:logistic",
    eval_metric="aucpr",
    scale_pos_weight=scale_pos_weight,
    random_state=RANDOM_STATE,
)

xgb.fit(X_train, y_train)
xgb_metrics = evaluate_model(xgb, X_validation, y_validation)
```

iv. LightGBM

```
lgbm = lgb.LGBMClassifier(
    objective="binary", scale_pos_weight=scale_pos_weight, random_state=RANDOM_STATE
)

lgbm.fit(X_train, y_train)
lgbm_metrics = evaluate_model(lgbm, X_validation, y_validation)
```

```
[LightGBM] [Info] Number of positive: 3385, number of negative: 25549
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.0014
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 176
[LightGBM] [Info] Number of data points in the train set: 28934, number of used features: 59
```

```
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.116990 -> initscore=-
2.021244
[LightGBM] [Info] Start training from score -2.021244
```

v. CatBoost

```python
cat = CatBoostClassifier(
    loss_function="Logloss",
    eval_metric="PRAUC",
    class_weights=[1, scale_pos_weight],
    verbose=0,
    random_state=RANDOM_STATE,
)

cat.fit(X_train, y_train)
cat_metrics = evaluate_model(cat, X_validation, y_validation)
```

Results comparison

```python
results = pd.DataFrame.from_dict(
    {
        "Logistic Regression": log_reg_metrics,
        "Random Forest": rf_metrics,
        "XGBoost": xgb_metrics,
        "LightGBM": lgbm_metrics,
        "CatBoost": cat_metrics,
    },
    orient="index",
)

results.sort_values(by="AUC_PR", ascending=False)
```

	AUC_PR	F1	Precision	Recall
LightGBM	0.445832	0.459201	0.362826	0.625296
CatBoost	0.442484	0.459831	0.368159	0.612293
Logistic Regression	0.411748	0.395126	0.287325	0.632388
XGBoost	0.407729	0.434281	0.347795	0.578014

	AUC_PR	F1	Precision	Recall
Random Forest	0.313136	0.275808	0.413712	0.206856

On the validation set, LightGBM and CatBoost exhibit very similar performance, with nearly identical F1 scores and only a marginal difference in AUC-PR (0.446 for LightGBM versus 0.442 for CatBoost). Both models clearly outperform the other architectures and appear well suited to handling the strong class imbalance present in the data.

LightGBM was selected as the primary model architecture because it achieves the highest AUC-PR overall, indicating slightly better ranking of likely subscribers, while maintaining strong recall and competitive precision. Given the minimal performance gap, this choice is pragmatic rather than definitive; CatBoost remains a strong alternative and could be expected to perform similarly in downstream analysis. **LightGBM** is therefore used for the remainder of the project to provide a consistent baseline for both global and segment-specific modeling.

2. Global model

Following the architecture selection, we now develop a fully optimized Global Model. This model serves as the primary benchmark for the project, trained on the full training dataset, containing observations from all clusters, to capture broad customer patterns. To manage the significant class imbalance (~11.7% success rate), we implement an analytical class-weighting strategy (scale_pos_weight) that penalizes errors on the minority class more heavily, ensuring the model prioritizes potential subscribers over the majority of non-subscribers.

Once we decided to work with LightGBM, we'll train the global model which will the observations from all clusters.

```
X_train_, y_train = load_dataset("02", "train")
X_validation_, y_validation = load_dataset("02", "validation")
```

```
train
X shape: (28934, 60)
y shape: (28934,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	0.0

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0

```
validation
X shape: (7234, 60)
y shape: (7234,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

```python
cluster_feat = "cluster"

X_train = X_train_.copy()
X_train = X_train.drop(columns=cluster_feat)
print("\nTrain set: ", X_train.shape)
display(X_train.head(3))

X_validation = X_validation_.copy()
X_validation = X_validation.drop(columns=cluster_feat).copy()
print("\nValidation set: ", X_validation.shape)
display(X_validation.head(3))
```

```
Train set:  (28934, 59)
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	0.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0

```
Validation set:  (7234, 59)
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

```
n_pos = y_train.sum()
n_neg = len(y_train) - n_pos
scale_pos_weight = n_neg / n_pos

print(f"Scale pos weight: {scale_pos_weight:.2f}")
```

```
Scale pos weight: 7.55
```

1. Hyperparameter Optimization via Optuna

We employ Bayesian Optimization using the Optuna framework to fine-tune the LightGBM hyperparameters. Unlike a standard grid search, this approach efficiently explores the parameter space to maximize the Area Under the Precision-Recall Curve (AUC-PR). This specific metric is chosen to ensure high model reliability and ranking performance within the minority positive class, which is essential for effective telemarketing targeting.

```
def objective(trial):
    params = {
        "objective": "binary",
        "metric": "aucpr",
        "random_state": RANDOM_STATE,
        "scale_pos_weight": scale_pos_weight,
        "verbosity": -1,
        # hyperparameters to tune
        "num_leaves": trial.suggest_int("num_leaves", 16, 128),
        "max_depth": trial.suggest_int("max_depth", -1, 12),
        "learning_rate": trial.suggest_float("learning_rate", 0.01, 0.2, log=True),
        "n_estimators": trial.suggest_int("n_estimators", 200, 600),
        "min_child_samples": trial.suggest_int("min_child_samples", 20, 200),
    }

    model = lgb.LGBMClassifier(**params)
    model.fit(X_train, y_train)

    y_val_prob = model.predict_proba(X_validation)[:, 1]
```

```
    aucpr = average_precision_score(y_validation, y_val_prob)

    return aucpr
```

```
study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=30, show_progress_bar=True)

print("Best AUC-PR:", study.best_value)
print("Best parameters:", study.best_params)

best_params = study.best_params
```

```
[I 2026-04-11 22:39:25,864] A new study created in memory with name: no-name-
a7d0f6a2-f999-450c-a217-c06396bf1f6f
Best trial: 0. Best value: 0.447814:   3%|          | 1/30 [00:02<01:07,  2.31s/it]

[I 2026-04-11 22:39:28,181] Trial 0 finished with value: 0.4478144657018903 and parameters: 

Best trial: 0. Best value: 0.447814:   7%|          | 2/30 [00:02<00:33,  1.21s/it]

[I 2026-04-11 22:39:28,609] Trial 1 finished with value: 0.4445655332491211 and parameters: 

Best trial: 0. Best value: 0.447814:  10%|          | 3/30 [00:11<02:05,  4.63s/it]

[I 2026-04-11 22:39:37,320] Trial 2 finished with value: 0.4391926050529471 and parameters: 
1, 'learning_rate': 0.01600332791287721, 'n_estimators': 394, 'min_child_samples': 187}. Bes

Best trial: 3. Best value: 0.456969:  13%|          | 4/30 [00:11<01:17,  2.96s/it]

[I 2026-04-11 22:39:37,730] Trial 3 finished with value: 0.45696934844549475 and parameters:

Best trial: 3. Best value: 0.456969:  17%|          | 5/30 [00:13<01:04,  2.56s/it]

[I 2026-04-11 22:39:39,581] Trial 4 finished with value: 0.44954968330392475 and parameters:

Best trial: 3. Best value: 0.456969:  20%|          | 6/30 [00:14<00:44,  1.87s/it]

[I 2026-04-11 22:39:40,116] Trial 5 finished with value: 0.44350602721399596 and parameters:
```

```
Best trial: 3. Best value: 0.456969:  23%|         | 7/30 [00:14<00:33,  1.47s/it]

[I 2026-04-11 22:39:40,764] Trial 6 finished with value: 0.44963846110177214 and parameters:

Best trial: 3. Best value: 0.456969:  27%|         | 8/30 [00:18<00:49,  2.27s/it]

[I 2026-04-11 22:39:44,732] Trial 7 finished with value: 0.4313467669736031 and parameters: 

Best trial: 3. Best value: 0.456969:  30%|         | 9/30 [00:21<00:52,  2.50s/it]

[I 2026-04-11 22:39:47,741] Trial 8 finished with value: 0.43094627551942083 and parameters:

Best trial: 3. Best value: 0.456969:  33%|        | 10/30 [00:22<00:40,  2.02s/it]

[I 2026-04-11 22:39:48,682] Trial 9 finished with value: 0.4526226413460322 and parameters: 

Best trial: 3. Best value: 0.456969:  37%|        | 11/30 [00:25<00:39,  2.08s/it]

[I 2026-04-11 22:39:50,899] Trial 10 finished with value: 0.3871406869432618 and parameters:

Best trial: 3. Best value: 0.456969:  40%|        | 12/30 [00:25<00:28,  1.60s/it]

[I 2026-04-11 22:39:51,392] Trial 11 finished with value: 0.4564347071532654 and parameters:

Best trial: 3. Best value: 0.456969:  43%|        | 13/30 [00:25<00:20,  1.22s/it]

[I 2026-04-11 22:39:51,746] Trial 12 finished with value: 0.4440400281233293 and parameters:

Best trial: 3. Best value: 0.456969:  47%|        | 14/30 [00:26<00:14,  1.07it/s]

[I 2026-04-11 22:39:52,019] Trial 13 finished with value: 0.4435291437650471 and parameters:

Best trial: 3. Best value: 0.456969:  53%|        | 16/30 [00:27<00:09,  1.50it/s]

[I 2026-04-11 22:39:52,748] Trial 14 finished with value: 0.43706673116171496 and parameters
[I 2026-04-11 22:39:52,942] Trial 15 finished with value: 0.40910326269498287 and parameters
```

```
Best trial: 3. Best value: 0.456969:  57%|       | 17/30 [00:28<00:11,  1.12it/s]

[I 2026-04-11 22:39:54,352] Trial 16 finished with value: 0.41873884512182197 and parameters

Best trial: 17. Best value: 0.458737:  60%|       | 18/30 [00:29<00:09,  1.25it/s]

[I 2026-04-11 22:39:54,935] Trial 17 finished with value: 0.45873686834678906 and parameters

Best trial: 17. Best value: 0.458737:  63%|       | 19/30 [00:29<00:08,  1.29it/s]

[I 2026-04-11 22:39:55,651] Trial 18 finished with value: 0.45708157305068425 and parameters

Best trial: 17. Best value: 0.458737:  67%|       | 20/30 [00:30<00:08,  1.20it/s]

[I 2026-04-11 22:39:56,628] Trial 19 finished with value: 0.4504615038099045 and parameters:

Best trial: 17. Best value: 0.458737:  70%|       | 21/30 [00:32<00:08,  1.01it/s]

[I 2026-04-11 22:39:57,981] Trial 20 finished with value: 0.42780850601559517 and parameters

Best trial: 17. Best value: 0.458737:  73%|       | 22/30 [00:32<00:06,  1.15it/s]

[I 2026-04-11 22:39:58,578] Trial 21 finished with value: 0.45658472013673357 and parameters

Best trial: 17. Best value: 0.458737:  77%|       | 23/30 [00:33<00:05,  1.23it/s]

[I 2026-04-11 22:39:59,243] Trial 22 finished with value: 0.4454243851992407 and parameters:

Best trial: 17. Best value: 0.458737:  80%|       | 24/30 [00:33<00:04,  1.44it/s]

[I 2026-04-11 22:39:59,665] Trial 23 finished with value: 0.44965349113864017 and parameters

Best trial: 17. Best value: 0.458737:  83%|       | 25/30 [00:35<00:04,  1.10it/s]

[I 2026-04-11 22:40:01,076] Trial 24 finished with value: 0.45204708320422404 and parameters

Best trial: 17. Best value: 0.458737:  87%|       | 26/30 [00:35<00:03,  1.30it/s]
```

```
[I 2026-04-11 22:40:01,512] Trial 25 finished with value: 0.4559581277073126 and parameters:

Best trial: 17. Best value: 0.458737:  90%|      | 27/30 [00:35<00:01,  1.67it/s]

[I 2026-04-11 22:40:01,716] Trial 26 finished with value: 0.3926110860738095 and parameters:

Best trial: 17. Best value: 0.458737:  93%|      | 28/30 [00:36<00:01,  1.58it/s]

[I 2026-04-11 22:40:02,424] Trial 27 finished with value: 0.4451601359776289 and parameters:

Best trial: 17. Best value: 0.458737:  97%|      | 29/30 [00:37<00:00,  1.28it/s]

[I 2026-04-11 22:40:03,549] Trial 28 finished with value: 0.44293438467538027 and parameters

Best trial: 17. Best value: 0.458737: 100%|      | 30/30 [00:39<00:00,  1.32s/it]

[I 2026-04-11 22:40:05,461] Trial 29 finished with value: 0.4487213187391542 and parameters:
Best AUC-PR: 0.45873686834678906
Best parameters: {'num_leaves': 56, 'max_depth': 4, 'learning_rate': 0.0667133655535366, 'n_e
```

We train the final global base model with the best parameters

```python
global_model = lgb.LGBMClassifier(
    objective="binary",
    scale_pos_weight=scale_pos_weight,
    random_state=RANDOM_STATE,
    **best_params
)

global_model.fit(X_train, y_train)
```

boosting_type	'gbdt'
num_leaves	56
max_depth	4
learning_rate	0.0667133655535366
n_estimators	363

subsample_for_bin	200000
objective	'binary'
class_weight	None
min_split_gain	0.0
min_child_weight	0.001
min_child_samples	51
subsample	1.0
subsample_freq	0
colsample_bytree	1.0
reg_alpha	0.0
reg_lambda	0.0
random_state	705
n_jobs	None
importance_type	'split'
scale_pos_weight	7.547710487444609

2. Probability Calibration

Machine learning models, particularly those using class-weighting, often produce "scores" rather than true probabilities. We apply Isotonic Regression to calibrate these outputs, aligning the model's predicted likelihoods with the actual observed subscription rates. Well-calibrated probabilities are a prerequisite for our upcoming Business Cost Framework, where we will calculate expected ROI based on specific call costs and subscription values.

```
# raw probabilities on validation
val_probs_raw = global_model.predict_proba(X_validation)[:, 1]

# isotonic regression calibration
calibrator = IsotonicRegression(out_of_bounds="clip")
calibrator.fit(val_probs_raw, y_validation)

# calibrated probabilities
val_probs_calibrated = calibrator.transform(val_probs_raw)
```

```
global_validation_metrics = evaluate_from_probs(y_validation, val_probs_calibrated)

print("Global model (validation, calibrated):")
print(global_validation_metrics)
```

```
Global model (validation, calibrated):
{'AUC_PR': 0.44694232107218884, 'F1': 0.31654676258992803, 'Precision': 0.6616541353383458,
```

3. Baseline Performance across Customer Segments

To evaluate the "one-size-fits-all" approach, we project the Global Model's performance onto our four distinct customer clusters. This diagnostic step identifies where the global patterns are robust and where they fail to capture localized nuances.

```
cluster_results = {}

for c in sorted(X_validation_[cluster_feat].unique()):
    idx = X_validation_[cluster_feat] == c

    cluster_results[f"cluster_{c}"] = evaluate_from_probs(
        y_validation.loc[idx], val_probs_calibrated[idx]
    )

cluster_results_df = pd.DataFrame(cluster_results).T
cluster_results_df
```

	AUC_PR	F1	Precision	Recall
cluster_0	0.573545	0.444444	0.769231	0.312500
cluster_1	0.292097	0.177340	0.562500	0.105263
cluster_2	0.373361	0.260504	0.574074	0.168478
cluster_3	0.501478	0.334165	0.656863	0.224080

```
model_dir = Path("models")

model_name = "global"

# save base model
joblib.dump(global_model, model_dir / f"{model_name}_model.joblib")

# save calibrator
joblib.dump(calibrator, model_dir / f"{model_name}_calibrator.joblib")

# save metadata
metadata = {
    "best_params": best_params,
    "scale_pos_weight": scale_pos_weight,
    "features": list(X_train.columns),
    "random_state": RANDOM_STATE,
}
```

```
joblib.dump(metadata, model_dir / f"{model_name}_metadata.joblib")
```

```
['models/global_metadata.joblib']
```

3. Cluster 0 model

```
cluster_id = 0
X_train, y_train = load_dataset(nb="02", split="train", cluster_id=cluster_id)
X_validation, y_validation = load_dataset(
    nb="02", split="validation", cluster_id=cluster_id
)
```

```
train | cluster 0
X shape: (5043, 59)
y shape: (5043,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
2	0.0	0.0	0.0	0.0	0.0
6	0.0	0.0	0.0	0.0	0.0
10	0.0	0.0	0.0	0.0	0.0

```
validation | cluster 0
X shape: (1271, 59)
y shape: (1271,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0
3	0.0	0.0	0.0	0.0	1.0

```python
# 2. Calculate the specific scale_pos_weight for Cluster 0
# Segment-specific weighting is essential as subscription rates vary by cluster
n_pos = y_train.sum()
n_neg = len(y_train) - n_pos
scale_pos_weight = n_neg / n_pos

print(f"Cluster {cluster_id} - Training Shape: {X_train.shape}")
print(f"Cluster {cluster_id} - Scale Pos Weight: {scale_pos_weight:.2f}")
```

```
Cluster 0 - Training Shape: (5043, 59)
Cluster 0 - Scale Pos Weight: 6.14
```

3.1. Hyperparameter Optimization via Optuna

```python
def objective(trial):
    params = {
        "objective": "binary",
        "metric": "aucpr",
        "random_state": RANDOM_STATE,
        "scale_pos_weight": scale_pos_weight,
        "verbosity": -1,
        # hyperparameters to tune
        "num_leaves": trial.suggest_int("num_leaves", 16, 128),
        "max_depth": trial.suggest_int("max_depth", -1, 12),
        "learning_rate": trial.suggest_float("learning_rate", 0.01, 0.2, log=True),
        "n_estimators": trial.suggest_int("n_estimators", 200, 600),
        "min_child_samples": trial.suggest_int("min_child_samples", 20, 200),
    }

    model = lgb.LGBMClassifier(**params)
    model.fit(X_train, y_train)

    y_val_prob = model.predict_proba(X_validation)[:, 1]
    aucpr = average_precision_score(y_validation, y_val_prob)

    return aucpr
```

```python
study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=30, show_progress_bar=True)
```

```python
print("Best AUC-PR:", study.best_value)
print("Best parameters:", study.best_params)

best_params = study.best_params
```

```
[I 2026-04-11 22:40:06,370] A new study created in memory with name: no-name-
b351166e-8b64-4358-93e2-007ada53965d
Best trial: 0. Best value: 0.531066:   3%|          | 1/30 [00:02<01:09,  2.39s/it]

[I 2026-04-11 22:40:08,763] Trial 0 finished with value: 0.5310661472319013 and parameters: {

Best trial: 0. Best value: 0.531066:   7%|          | 2/30 [00:02<00:36,  1.30s/it]

[I 2026-04-11 22:40:09,291] Trial 1 finished with value: 0.5244108585551055 and parameters: {

Best trial: 2. Best value: 0.538923:  10%|          | 3/30 [00:03<00:29,  1.08s/it]

[I 2026-04-11 22:40:10,117] Trial 2 finished with value: 0.5389225910710985 and parameters: {

Best trial: 3. Best value: 0.547976:  13%|          | 4/30 [00:03<00:19,  1.35it/s]

[I 2026-04-11 22:40:10,334] Trial 3 finished with value: 0.5479760727337629 and parameters: {

Best trial: 3. Best value: 0.547976:  17%|          | 5/30 [00:05<00:21,  1.14it/s]

[I 2026-04-11 22:40:11,450] Trial 4 finished with value: 0.5353698136940971 and parameters: {
1, 'learning_rate': 0.010797614566439385, 'n_estimators': 253, 'min_child_samples': 125}. Bes

Best trial: 3. Best value: 0.547976:  20%|          | 6/30 [00:06<00:22,  1.09it/s]

[I 2026-04-11 22:40:12,452] Trial 5 finished with value: 0.5322610168288402 and parameters: {

Best trial: 3. Best value: 0.547976:  23%|          | 7/30 [00:10<00:43,  1.91s/it]

[I 2026-04-11 22:40:16,389] Trial 6 finished with value: 0.5236354925663569 and parameters: {
1, 'learning_rate': 0.04085870153679321, 'n_estimators': 544, 'min_child_samples': 94}. Best
```

```
Best trial: 3. Best value: 0.547976:  27%|         | 8/30 [00:11<00:38,  1.73s/it]

[I 2026-04-11 22:40:17,750] Trial 7 finished with value: 0.5298355297600893 and parameters: {

Best trial: 3. Best value: 0.547976:  30%|         | 9/30 [00:12<00:32,  1.54s/it]

[I 2026-04-11 22:40:18,854] Trial 8 finished with value: 0.5457747905264096 and parameters: {

Best trial: 3. Best value: 0.547976:  33%|        | 10/30 [00:13<00:28,  1.43s/it]

[I 2026-04-11 22:40:20,047] Trial 9 finished with value: 0.5399970524933088 and parameters: {

Best trial: 3. Best value: 0.547976:  37%|        | 11/30 [00:14<00:22,  1.21s/it]

[I 2026-04-11 22:40:20,749] Trial 10 finished with value: 0.520270351459665 and parameters: {

Best trial: 11. Best value: 0.549778:  40%|        | 12/30 [00:14<00:16,  1.09it/s]

[I 2026-04-11 22:40:21,015] Trial 11 finished with value: 0.5497779357150551 and parameters:

Best trial: 11. Best value: 0.549778:  47%|        | 14/30 [00:15<00:08,  1.82it/s]

[I 2026-04-11 22:40:21,258] Trial 12 finished with value: 0.5289717912241562 and parameters:
[I 2026-04-11 22:40:21,424] Trial 13 finished with value: 0.545340839809516 and parameters: {

Best trial: 11. Best value: 0.549778:  50%|        | 15/30 [00:15<00:06,  2.24it/s]

[I 2026-04-11 22:40:21,635] Trial 14 finished with value: 0.5420670727842102 and parameters:

Best trial: 11. Best value: 0.549778:  57%|        | 17/30 [00:16<00:06,  1.95it/s]

[I 2026-04-11 22:40:22,786] Trial 15 finished with value: 0.47701856479372573 and parameters
[I 2026-04-11 22:40:22,962] Trial 16 finished with value: 0.5478407356502513 and parameters:

Best trial: 11. Best value: 0.549778:  60%|        | 18/30 [00:17<00:07,  1.71it/s]

[I 2026-04-11 22:40:23,716] Trial 17 finished with value: 0.45847090222522724 and parameters
```

```
Best trial: 11. Best value: 0.549778:  63%|       | 19/30 [00:18<00:08,  1.24it/s]

[I 2026-04-11 22:40:25,037] Trial 18 finished with value: 0.49618050785877743 and parameters

Best trial: 11. Best value: 0.549778:  70%|       | 21/30 [00:19<00:04,  1.91it/s]

[I 2026-04-11 22:40:25,432] Trial 19 finished with value: 0.5285135455534031 and parameters:
[I 2026-04-11 22:40:25,583] Trial 20 finished with value: 0.528544291153175 and parameters: 

Best trial: 11. Best value: 0.549778:  73%|       | 22/30 [00:19<00:03,  2.37it/s]

[I 2026-04-11 22:40:25,767] Trial 21 finished with value: 0.5466138100921406 and parameters:

Best trial: 11. Best value: 0.549778:  77%|       | 23/30 [00:19<00:02,  2.56it/s]

[I 2026-04-11 22:40:26,083] Trial 22 finished with value: 0.5270059445125919 and parameters:
[I 2026-04-11 22:40:26,179] Trial 23 finished with value: 0.5201204665035678 and parameters:

Best trial: 11. Best value: 0.549778:  83%|       | 25/30 [00:20<00:01,  3.16it/s]

[I 2026-04-11 22:40:26,544] Trial 24 finished with value: 0.5370912503476131 and parameters:

Best trial: 11. Best value: 0.549778:  87%|       | 26/30 [00:20<00:01,  3.28it/s]

[I 2026-04-11 22:40:26,814] Trial 25 finished with value: 0.5453907561562835 and parameters:

Best trial: 11. Best value: 0.549778:  93%|       | 28/30 [00:21<00:00,  3.17it/s]

[I 2026-04-11 22:40:27,403] Trial 26 finished with value: 0.5411352756934713 and parameters:
[I 2026-04-11 22:40:27,547] Trial 27 finished with value: 0.5328586176555753 and parameters:

Best trial: 11. Best value: 0.549778:  97%|       | 29/30 [00:21<00:00,  2.78it/s]

[I 2026-04-11 22:40:28,021] Trial 28 finished with value: 0.5321511606368267 and parameters:

Best trial: 11. Best value: 0.549778: 100%|       | 30/30 [00:21<00:00,  1.37it/s]
```

```
[I 2026-04-11 22:40:28,244] Trial 29 finished with value: 0.5456495419258414 and parameters:
Best AUC-PR: 0.5497779357150551
Best parameters: {'num_leaves': 97, 'max_depth': 3, 'learning_rate': 0.12574149490345382, 'n_
```

```python
model = lgb.LGBMClassifier(
    objective="binary",
    scale_pos_weight=scale_pos_weight,
    random_state=RANDOM_STATE,
    **best_params
)

model.fit(X_train, y_train)
```

boosting_type	'gbdt'
num_leaves	97
max_depth	3
learning_rate	0.12574149490345382
n_estimators	358
subsample_for_bin	200000
objective	'binary'
class_weight	None
min_split_gain	0.0
min_child_weight	0.001
min_child_samples	51
subsample	1.0
subsample_freq	0
colsample_bytree	1.0
reg_alpha	0.0
reg_lambda	0.0
random_state	705
n_jobs	None
importance_type	'split'
scale_pos_weight	6.143059490084986

3.2. Probability Calibration

```
# raw probabilities on validation
val_probs_raw = model.predict_proba(X_validation)[:, 1]

# isotonic regression calibration
calibrator = IsotonicRegression(out_of_bounds="clip")
calibrator.fit(val_probs_raw, y_validation)

# calibrated probabilities
val_probs_calibrated = calibrator.transform(val_probs_raw)
```

```
c0_validation_metrics = evaluate_from_probs(y_validation, val_probs_calibrated)

print(f"Cluster {cluster_id} model (validation, calibrated):")
print(c0_validation_metrics)
```

```
Cluster 0 model (validation, calibrated):
{'AUC_PR': 0.5456038645178486, 'F1': 0.4460431654676259, 'Precision': 0.7209302325581395, 'Re
```

We save the model

```
model_dir = Path("models")

model_name = f"c{cluster_id}"

# save base model
joblib.dump(model, model_dir / f"{model_name}_model.joblib")

# save calibrator
joblib.dump(calibrator, model_dir / f"{model_name}_calibrator.joblib")

# save metadata
metadata = {
    "best_params": best_params,
    "scale_pos_weight": scale_pos_weight,
    "features": list(X_train.columns),
    "random_state": RANDOM_STATE,
}

joblib.dump(metadata, model_dir / f"{model_name}_metadata.joblib")
```

```
['models/c0_metadata.joblib']
```

4. Cluster 1 model

```
cluster_id = 1
X_train, y_train = load_dataset(nb="02", split="train", cluster_id=cluster_id)
X_validation, y_validation = load_dataset(
    nb="02", split="validation", cluster_id=cluster_id
)
```

```
train | cluster 1
X shape: (8847, 59)
y shape: (8847,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	0.0
4	1.0	0.0	0.0	0.0	0.0
5	0.0	0.0	0.0	0.0	0.0

```
validation | cluster 1
X shape: (2266, 59)
y shape: (2266,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
5	0.0	0.0	0.0	0.0	0.0
15	1.0	0.0	0.0	0.0	0.0
16	0.0	0.0	0.0	0.0	0.0

```
test | cluster 1
X shape: (2769, 59)
y shape: (2769,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
3	0.0	0.0	0.0	0.0	0.0
4	0.0	1.0	0.0	0.0	0.0
6	0.0	1.0	0.0	0.0	0.0

```
# 2. Calculate the specific scale_pos_weight for Cluster 1
n_pos = y_train.sum()
n_neg = len(y_train) - n_pos
scale_pos_weight = n_neg / n_pos

print(f"Cluster {cluster_id} - Training Shape: {X_train.shape}")
print(f"Cluster {cluster_id} - Scale Pos Weight: {scale_pos_weight:.2f}")
```

```
Cluster 1 - Training Shape: (8847, 59)
Cluster 1 - Scale Pos Weight: 11.34
```

4.1. Hyperparameter Optimization via Optuna

```
def objective(trial):
    params = {
        "objective": "binary",
        "metric": "aucpr",
        "random_state": RANDOM_STATE,
        "scale_pos_weight": scale_pos_weight,
        "verbosity": -1,
        "num_leaves": trial.suggest_int("num_leaves", 16, 128),
        "max_depth": trial.suggest_int("max_depth", -1, 12),
        "learning_rate": trial.suggest_float("learning_rate", 0.01, 0.2, log=True),
        "n_estimators": trial.suggest_int("n_estimators", 200, 600),
        "min_child_samples": trial.suggest_int("min_child_samples", 20, 200),
    }

    model = lgb.LGBMClassifier(**params)
    model.fit(X_train, y_train)

    y_val_prob = model.predict_proba(X_validation)[:, 1]
    aucpr = average_precision_score(y_validation, y_val_prob)
    return aucpr
```

```python
study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=30, show_progress_bar=True)

print("Best AUC-PR:", study.best_value)
print("Best parameters:", study.best_params)

best_params = study.best_params
```

```
[I 2026-04-11 22:40:29,205] A new study created in memory with name: no-name-
8fc3d20f-c078-4490-a2a2-45f069462384
Best trial: 0. Best value: 0.257394:   3%|          | 1/30 [00:00<00:21,  1.32it/s]

[I 2026-04-11 22:40:29,968] Trial 0 finished with value: 0.25739362028779494 and parameters:

Best trial: 0. Best value: 0.257394:   7%|          | 2/30 [00:03<00:51,  1.85s/it]

[I 2026-04-11 22:40:32,585] Trial 1 finished with value: 0.222551034181087 and parameters: {

Best trial: 0. Best value: 0.257394:  10%|          | 3/30 [00:06<01:04,  2.39s/it]

[I 2026-04-11 22:40:35,615] Trial 2 finished with value: 0.22669006416987508 and parameters:

Best trial: 0. Best value: 0.257394:  13%|          | 4/30 [00:08<00:58,  2.24s/it]

[I 2026-04-11 22:40:37,635] Trial 3 finished with value: 0.21055585442185534 and parameters:

Best trial: 0. Best value: 0.257394:  17%|          | 5/30 [00:08<00:39,  1.57s/it]

[I 2026-04-11 22:40:38,024] Trial 4 finished with value: 0.25489154773760886 and parameters:

Best trial: 0. Best value: 0.257394:  20%|          | 6/30 [00:09<00:28,  1.19s/it]

[I 2026-04-11 22:40:38,475] Trial 5 finished with value: 0.22376589989646134 and parameters:

Best trial: 0. Best value: 0.257394:  23%|          | 7/30 [00:10<00:27,  1.19s/it]

[I 2026-04-11 22:40:39,673] Trial 6 finished with value: 0.24891885276705658 and parameters:
```

```
Best trial: 7. Best value: 0.283768:  27%|         | 8/30 [00:11<00:25,  1.14s/it]

[I 2026-04-11 22:40:40,709] Trial 7 finished with value: 0.2837675880425302 and parameters: {

Best trial: 7. Best value: 0.283768:  30%|         | 9/30 [00:13<00:28,  1.34s/it]

[I 2026-04-11 22:40:42,482] Trial 8 finished with value: 0.23847366606496317 and parameters:

Best trial: 7. Best value: 0.283768:  33%|        | 10/30 [00:16<00:39,  1.97s/it]

[I 2026-04-11 22:40:45,867] Trial 9 finished with value: 0.208112907271905 and parameters: {

Best trial: 7. Best value: 0.283768:  37%|        | 11/30 [00:16<00:27,  1.46s/it]

[I 2026-04-11 22:40:46,175] Trial 10 finished with value: 0.24045545064071322 and parameters

Best trial: 7. Best value: 0.283768:  40%|        | 12/30 [00:17<00:21,  1.19s/it]

[I 2026-04-11 22:40:46,730] Trial 11 finished with value: 0.26196511162518815 and parameters

Best trial: 7. Best value: 0.283768:  43%|        | 13/30 [00:18<00:16,  1.01it/s]

[I 2026-04-11 22:40:47,284] Trial 12 finished with value: 0.2653143124404377 and parameters:

Best trial: 7. Best value: 0.283768:  47%|        | 14/30 [00:18<00:14,  1.09it/s]

[I 2026-04-11 22:40:48,024] Trial 13 finished with value: 0.26403417383486344 and parameters

Best trial: 7. Best value: 0.283768:  50%|        | 15/30 [00:19<00:12,  1.24it/s]

[I 2026-04-11 22:40:48,571] Trial 14 finished with value: 0.23273635907137508 and parameters

Best trial: 7. Best value: 0.283768:  53%|        | 16/30 [00:20<00:11,  1.25it/s]

[I 2026-04-11 22:40:49,364] Trial 15 finished with value: 0.27633647953123847 and parameters

Best trial: 7. Best value: 0.283768:  57%|        | 17/30 [00:20<00:08,  1.47it/s]
```

```
[I 2026-04-11 22:40:49,763] Trial 16 finished with value: 0.2617813552136476 and parameters:

Best trial: 7. Best value: 0.283768:  60%|       | 18/30 [00:21<00:08,  1.42it/s]

[I 2026-04-11 22:40:50,521] Trial 17 finished with value: 0.2577897075211459 and parameters:

Best trial: 7. Best value: 0.283768:  63%|       | 19/30 [00:22<00:08,  1.29it/s]

[I 2026-04-11 22:40:51,453] Trial 18 finished with value: 0.2664669389986381 and parameters:

Best trial: 7. Best value: 0.283768:  67%|       | 20/30 [00:23<00:07,  1.28it/s]

[I 2026-04-11 22:40:52,260] Trial 19 finished with value: 0.2608809190292961 and parameters:

Best trial: 7. Best value: 0.283768:  70%|       | 21/30 [00:24<00:07,  1.16it/s]

[I 2026-04-11 22:40:53,316] Trial 20 finished with value: 0.2567628792879916 and parameters:

Best trial: 7. Best value: 0.283768:  73%|      | 22/30 [00:25<00:07,  1.11it/s]

[I 2026-04-11 22:40:54,292] Trial 21 finished with value: 0.26195063916483935 and parameters

Best trial: 7. Best value: 0.283768:  77%|      | 23/30 [00:26<00:06,  1.01it/s]

[I 2026-04-11 22:40:55,499] Trial 22 finished with value: 0.2625636434368555 and parameters:

Best trial: 7. Best value: 0.283768:  80%|      | 24/30 [00:27<00:06,  1.15s/it]

[I 2026-04-11 22:40:57,004] Trial 23 finished with value: 0.2708977859932459 and parameters:

Best trial: 7. Best value: 0.283768:  83%|      | 25/30 [00:28<00:05,  1.07s/it]

[I 2026-04-11 22:40:57,887] Trial 24 finished with value: 0.26554914176540007 and parameters

Best trial: 7. Best value: 0.283768:  87%|      | 26/30 [00:29<00:04,  1.14s/it]

[I 2026-04-11 22:40:59,202] Trial 25 finished with value: 0.2645417005559937 and parameters:
```

```
Best trial: 7. Best value: 0.283768:  90%|      | 27/30 [00:31<00:03,  1.26s/it]

[I 2026-04-11 22:41:00,723] Trial 26 finished with value: 0.28270172438092284 and parameters

Best trial: 7. Best value: 0.283768:  93%|      | 28/30 [00:32<00:02,  1.18s/it]

[I 2026-04-11 22:41:01,728] Trial 27 finished with value: 0.27360415680660805 and parameters

Best trial: 7. Best value: 0.283768:  97%|      | 29/30 [00:33<00:01,  1.11s/it]

[I 2026-04-11 22:41:02,684] Trial 28 finished with value: 0.26877437221327594 and parameters

Best trial: 7. Best value: 0.283768: 100%|      | 30/30 [00:34<00:00,  1.16s/it]

[I 2026-04-11 22:41:03,873] Trial 29 finished with value: 0.2649295466527023 and parameters:
Best AUC-PR: 0.2837675880425302
Best parameters: {'num_leaves': 61, 'max_depth': 10, 'learning_rate': 0.05352170002108291, '
```

```python
model = lgb.LGBMClassifier(
    objective="binary",
    scale_pos_weight=scale_pos_weight,
    random_state=RANDOM_STATE,
    **best_params
)

model.fit(X_train, y_train)
```

boosting_type	'gbdt'
num_leaves	61
max_depth	10
learning_rate	0.05352170002108291
n_estimators	257
subsample_for_bin	200000
objective	'binary'
class_weight	None
min_split_gain	0.0

min_child_weight	0.001
min_child_samples	125
subsample	1.0
subsample_freq	0
colsample_bytree	1.0
reg_alpha	0.0
reg_lambda	0.0
random_state	705
n_jobs	None
importance_type	'split'
scale_pos_weight	11.338912133891213

4.2. Probability Calibration

```
# raw probabilities on validation
val_probs_raw = model.predict_proba(X_validation)[:, 1]

# isotonic regression calibration
calibrator = IsotonicRegression(out_of_bounds="clip")
calibrator.fit(val_probs_raw, y_validation)

# calibrated probabilities
val_probs_calibrated = calibrator.transform(val_probs_raw)
```

```
c1_validation_metrics = evaluate_from_probs(y_validation, val_probs_calibrated)

print(f"Cluster {cluster_id} model (validation, calibrated):")
print(c1_validation_metrics)
```

```
Cluster 1 model (validation, calibrated):
{'AUC_PR': 0.2786453337479804, 'F1': 0.19900497512437812, 'Precision': 0.6666666666666666, '
```

```
# save the model and calibrator for cluster 1
model_dir = Path("models")
model_name = f"c{cluster_id}"

# save model
joblib.dump(model, model_dir / f"{model_name}_model.joblib")
```

```python
# save calibrator
joblib.dump(calibrator, model_dir / f"{model_name}_calibrator.joblib")

# save metadata
metadata = {
    "best_params": best_params,
    "scale_pos_weight": scale_pos_weight,
    "features": list(X_train.columns),
    "random_state": RANDOM_STATE,
}

joblib.dump(metadata, model_dir / f"{model_name}_metadata.joblib")
```

```
['models/c1_metadata.joblib']
```

5. Cluster 2 model

```python
cluster_id = 2
X_train, y_train = load_dataset(nb="02", split="train", cluster_id=cluster_id)
X_validation, y_validation = load_dataset(
    nb="02", split="validation", cluster_id=cluster_id
)
```

```
train | cluster 2
X shape: (7058, 59)
y shape: (7058,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
1	0.0	1.0	0.0	0.0	0.0
11	0.0	1.0	0.0	0.0	0.0
12	0.0	0.0	0.0	0.0	0.0

```
validation | cluster 2
X shape: (1720, 59)
```

```
y shape: (1720,)

X head:
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__job_
8	0.0	0.0	0.0	0.0	0.0
10	0.0	0.0	0.0	0.0	0.0
11	0.0	0.0	0.0	0.0	0.0

```
test | cluster 2
X shape: (2201, 59)
y shape: (2201,)

X head:
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__job_
1	0.0	1.0	0.0	0.0	0.0
5	1.0	0.0	0.0	0.0	0.0
13	0.0	1.0	0.0	0.0	0.0

```python
# 2. Calculate the specific scale_pos_weight for Cluster 2
n_pos = y_train.sum()
n_neg = len(y_train) - n_pos
scale_pos_weight = n_neg / n_pos

print(f"Cluster {cluster_id} - Training Shape: {X_train.shape}")
print(f"Cluster {cluster_id} - Scale Pos Weight: {scale_pos_weight:.2f}")
```

```
Cluster 2 - Training Shape: (7058, 59)
Cluster 2 - Scale Pos Weight: 7.96
```

5.1. Hyperparameter Optimization via Optuna

```python
def objective(trial):
    params = {
        "objective": "binary",
        "metric": "aucpr",
        "random_state": RANDOM_STATE,
        "scale_pos_weight": scale_pos_weight,
        "verbosity": -1,
        "num_leaves": trial.suggest_int("num_leaves", 16, 128),
        "max_depth": trial.suggest_int("max_depth", -1, 12),
        "learning_rate": trial.suggest_float("learning_rate", 0.01, 0.2, log=True),
        "n_estimators": trial.suggest_int("n_estimators", 200, 600),
        "min_child_samples": trial.suggest_int("min_child_samples", 20, 200),
    }

    model = lgb.LGBMClassifier(**params)
    model.fit(X_train, y_train)

    y_val_prob = model.predict_proba(X_validation)[:, 1]
    aucpr = average_precision_score(y_validation, y_val_prob)
    return aucpr
```

```python
study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=30, show_progress_bar=True)

print("Best AUC-PR:", study.best_value)
print("Best parameters:", study.best_params)

best_params = study.best_params
```

```
[I 2026-04-11 22:41:05,206] A new study created in memory with name: no-name-
601ff287-4a3a-4541-9ae9-da7d6cdc99ea
Best trial: 0. Best value: 0.34768:   3%|          | 1/30 [00:00<00:14,  2.01it/s]

[I 2026-04-11 22:41:05,710] Trial 0 finished with value: 0.3476797615973629 and parameters: 

Best trial: 0. Best value: 0.34768:   7%|          | 2/30 [00:01<00:18,  1.50it/s]

[I 2026-04-11 22:41:06,492] Trial 1 finished with value: 0.28922831072947786 and parameters:

Best trial: 0. Best value: 0.34768:  10%|          | 3/30 [00:02<00:28,  1.06s/it]
```

```
[I 2026-04-11 22:41:08,026] Trial 2 finished with value: 0.26943942994365316 and parameters:

Best trial: 0. Best value: 0.34768:  13%|         | 4/30 [00:03<00:26,  1.01s/it]

[I 2026-04-11 22:41:08,958] Trial 3 finished with value: 0.3141035173716834 and parameters: {

Best trial: 0. Best value: 0.34768:  17%|         | 5/30 [00:04<00:20,  1.19it/s]

[I 2026-04-11 22:41:09,493] Trial 4 finished with value: 0.25798959887157535 and parameters:

Best trial: 0. Best value: 0.34768:  20%|         | 6/30 [00:05<00:26,  1.09s/it]

[I 2026-04-11 22:41:11,057] Trial 5 finished with value: 0.331340450332198 and parameters: {

Best trial: 0. Best value: 0.34768:  23%|         | 7/30 [00:07<00:26,  1.17s/it]

[I 2026-04-11 22:41:12,403] Trial 6 finished with value: 0.2483694764718477 and parameters: {

Best trial: 0. Best value: 0.34768:  27%|         | 8/30 [00:08<00:25,  1.18s/it]

[I 2026-04-11 22:41:13,603] Trial 7 finished with value: 0.3299175430139323 and parameters: {

Best trial: 0. Best value: 0.34768:  30%|         | 9/30 [00:09<00:24,  1.19s/it]

[I 2026-04-11 22:41:14,815] Trial 8 finished with value: 0.30248672724165826 and parameters:
1, 'learning_rate': 0.01442468671601579, 'n_estimators': 429, 'min_child_samples': 196}. Best

Best trial: 0. Best value: 0.34768:  33%|        | 10/30 [00:10<00:19,  1.05it/s]

[I 2026-04-11 22:41:15,242] Trial 9 finished with value: 0.3310454469969295 and parameters: {

Best trial: 10. Best value: 0.351341:  37%|        | 11/30 [00:10<00:16,  1.16it/s]

[I 2026-04-11 22:41:15,887] Trial 10 finished with value: 0.3513412549800843 and parameters:

Best trial: 11. Best value: 0.352604:  40%|        | 12/30 [00:11<00:14,  1.28it/s]
```

```
[I 2026-04-11 22:41:16,490] Trial 11 finished with value: 0.352603720145778 and parameters: {

Best trial: 11. Best value: 0.352604:  47%|        | 14/30 [00:12<00:09,  1.75it/s]

[I 2026-04-11 22:41:17,240] Trial 12 finished with value: 0.33861628334109173 and parameters
1, 'learning_rate': 0.04759320505880708, 'n_estimators': 310, 'min_child_samples': 20}. Best
[I 2026-04-11 22:41:17,349] Trial 13 finished with value: 0.32444320157968787 and parameters

Best trial: 11. Best value: 0.352604:  53%|        | 16/30 [00:12<00:04,  2.97it/s]

[I 2026-04-11 22:41:17,456] Trial 14 finished with value: 0.33860895838261035 and parameters
[I 2026-04-11 22:41:17,572] Trial 15 finished with value: 0.33562140427528553 and parameters

Best trial: 17. Best value: 0.356615:  60%|        | 18/30 [00:12<00:03,  3.19it/s]

[I 2026-04-11 22:41:18,020] Trial 16 finished with value: 0.3423461840192134 and parameters:
[I 2026-04-11 22:41:18,202] Trial 17 finished with value: 0.35661459689099806 and parameters

Best trial: 17. Best value: 0.356615:  63%|        | 19/30 [00:13<00:03,  3.29it/s]

[I 2026-04-11 22:41:18,484] Trial 18 finished with value: 0.3348647011485238 and parameters:

Best trial: 17. Best value: 0.356615:  67%|        | 20/30 [00:14<00:04,  2.12it/s]

[I 2026-04-11 22:41:19,346] Trial 19 finished with value: 0.3094107415647866 and parameters:

Best trial: 17. Best value: 0.356615:  70%|        | 21/30 [00:14<00:03,  2.53it/s]

[I 2026-04-11 22:41:19,560] Trial 20 finished with value: 0.3496228125815707 and parameters:

Best trial: 17. Best value: 0.356615:  73%|        | 22/30 [00:15<00:04,  1.77it/s]

[I 2026-04-11 22:41:20,526] Trial 21 finished with value: 0.3420327262460883 and parameters:

Best trial: 17. Best value: 0.356615:  77%|        | 23/30 [00:17<00:07,  1.10s/it]

[I 2026-04-11 22:41:22,878] Trial 22 finished with value: 0.29291223037129743 and parameters
1, 'learning_rate': 0.05528153059102717, 'n_estimators': 341, 'min_child_samples': 28}. Best
```

```
Best trial: 17. Best value: 0.356615:  83%|      | 25/30 [00:19<00:04,  1.16it/s]

[I 2026-04-11 22:41:24,190] Trial 23 finished with value: 0.3270462949728678 and parameters:
[I 2026-04-11 22:41:24,351] Trial 24 finished with value: 0.335376748958731 and parameters: 

Best trial: 17. Best value: 0.356615:  87%|      | 26/30 [00:19<00:03,  1.33it/s]

[I 2026-04-11 22:41:24,841] Trial 25 finished with value: 0.3294888621670784 and parameters:

Best trial: 17. Best value: 0.356615:  93%|      | 28/30 [00:22<00:02,  1.02s/it]

[I 2026-04-11 22:41:27,670] Trial 26 finished with value: 0.2637064590606037 and parameters:
[I 2026-04-11 22:41:27,850] Trial 27 finished with value: 0.33901656209095055 and parameters

Best trial: 17. Best value: 0.356615:  97%|      | 29/30 [00:24<00:01,  1.24s/it]

[I 2026-04-11 22:41:29,603] Trial 28 finished with value: 0.28717595513263094 and parameters

Best trial: 17. Best value: 0.356615: 100%|      | 30/30 [00:24<00:00,  1.20it/s]

[I 2026-04-11 22:41:30,204] Trial 29 finished with value: 0.3533234986941247 and parameters:
Best AUC-PR: 0.35661459689099806
Best parameters: {'num_leaves': 54, 'max_depth': 2, 'learning_rate': 0.07657553475959958, 'n
```

```python
model = lgb.LGBMClassifier(
    objective="binary",
    scale_pos_weight=scale_pos_weight,
    random_state=RANDOM_STATE,
    **best_params
)

model.fit(X_train, y_train)
```

boosting_type	'gbdt'
num_leaves	54
max_depth	2
learning_rate	0.07657553475959958
n_estimators	368
subsample_for_bin	200000
objective	'binary'
class_weight	None
min_split_gain	0.0
min_child_weight	0.001
min_child_samples	55
subsample	1.0
subsample_freq	0
colsample_bytree	1.0
reg_alpha	0.0
reg_lambda	0.0
random_state	705
n_jobs	None
importance_type	'split'
scale_pos_weight	7.956852791878172

5.2. Probability Calibration

```
# raw probabilities on validation
val_probs_raw = model.predict_proba(X_validation)[:, 1]

# isotonic regression calibration
calibrator = IsotonicRegression(out_of_bounds="clip")
calibrator.fit(val_probs_raw, y_validation)

# calibrated probabilities
val_probs_calibrated = calibrator.transform(val_probs_raw)
```

```
c2_validation_metrics = evaluate_from_probs(y_validation, val_probs_calibrated)

print(f"Cluster {cluster_id} model (validation, calibrated):")
print(c2_validation_metrics)
```

```
Cluster 2 model (validation, calibrated):
{'AUC_PR': 0.3487701483135698, 'F1': 0.26609442060085836, 'Precision': 0.6326530612244898, '
```

```python
# save the model and calibrator for cluster 2
model_dir = Path("models")
model_name = f"c{cluster_id}"

# save model
joblib.dump(model, model_dir / f"{model_name}_model.joblib")

# save calibrator
joblib.dump(calibrator, model_dir / f"{model_name}_calibrator.joblib")

# save metadata
metadata = {
    "best_params": best_params,
    "scale_pos_weight": scale_pos_weight,
    "features": list(X_train.columns),
    "random_state": RANDOM_STATE,
}

joblib.dump(metadata, model_dir / f"{model_name}_metadata.joblib")
```

```
['models/c2_metadata.joblib']
```

6. Cluster 3 model

```python
cluster_id = 3
X_train, y_train = load_dataset(nb="02", split="train", cluster_id=cluster_id)
X_validation, y_validation = load_dataset(
    nb="02", split="validation", cluster_id=cluster_id
)
```

```
train | cluster 3
X shape: (7986, 59)
y shape: (7986,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
3	0.0	0.0	0.0	0.0	0.0
8	1.0	0.0	0.0	0.0	0.0
9	0.0	0.0	0.0	0.0	0.0

```
validation | cluster 3
X shape: (1977, 59)
y shape: (1977,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
6	0.0	0.0	1.0	0.0	0.0
12	0.0	0.0	0.0	0.0	0.0

```
test | cluster 3
X shape: (2461, 59)
y shape: (2461,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
8	0.0	0.0	0.0	0.0	0.0
9	0.0	0.0	0.0	0.0	1.0

```python
# Calculate cluster-specific scale_pos_weight
n_pos = y_train.sum()
n_neg = len(y_train) - n_pos
scale_pos_weight = n_neg / n_pos

print(f"Cluster {cluster_id} - Training Shape: {X_train.shape}")
print(f"Cluster {cluster_id} - Scale Pos Weight: {scale_pos_weight:.2f}")
```

```
Cluster 3 - Training Shape: (7986, 59)
Cluster 3 - Scale Pos Weight: 5.80
```

6.1. Hyperparameter Optimization via Optuna

```
def objective(trial):
    params = {
        "objective": "binary",
        "metric": "aucpr",
        "random_state": RANDOM_STATE,
        "scale_pos_weight": scale_pos_weight,
        "verbosity": -1,
        "num_leaves": trial.suggest_int("num_leaves", 16, 128),
        "max_depth": trial.suggest_int("max_depth", -1, 12),
        "learning_rate": trial.suggest_float("learning_rate", 0.01, 0.2, log=True),
        "n_estimators": trial.suggest_int("n_estimators", 200, 600),
        "min_child_samples": trial.suggest_int("min_child_samples", 20, 200),
    }

    model = lgb.LGBMClassifier(**params)
    model.fit(X_train, y_train)

    y_val_prob = model.predict_proba(X_validation)[:, 1]
    aucpr = average_precision_score(y_validation, y_val_prob)
    return aucpr
```

```
study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=30, show_progress_bar=True)

print("Best AUC-PR:", study.best_value)
print("Best parameters:", study.best_params)

best_params = study.best_params
```

```
[I 2026-04-11 22:41:30,794] A new study created in memory with name: no-name-
8427a1ae-77c7-48b3-9ad3-0d44f7fc4b30
Best trial: 0. Best value: 0.468725:   3%|          | 1/30 [00:04<01:58,  4.08s/it]

[I 2026-04-11 22:41:34,869] Trial 0 finished with value: 0.4687249958013377 and parameters: 

Best trial: 1. Best value: 0.490409:   7%|          | 2/30 [00:04<00:57,  2.06s/it]

[I 2026-04-11 22:41:35,518] Trial 1 finished with value: 0.49040930050961085 and parameters:
```

```
Best trial: 1. Best value: 0.490409:  10%|          | 3/30 [00:07<01:02,  2.31s/it]

[I 2026-04-11 22:41:38,131] Trial 2 finished with value: 0.43521339428702427 and parameters:

Best trial: 1. Best value: 0.490409:  13%|          | 4/30 [00:07<00:41,  1.59s/it]

[I 2026-04-11 22:41:38,628] Trial 3 finished with value: 0.4835536153117025 and parameters: {

Best trial: 4. Best value: 0.491037:  17%|          | 5/30 [00:08<00:30,  1.23s/it]

[I 2026-04-11 22:41:39,203] Trial 4 finished with value: 0.4910372467810263 and parameters: {

Best trial: 4. Best value: 0.491037:  20%|          | 6/30 [00:09<00:29,  1.22s/it]

[I 2026-04-11 22:41:40,412] Trial 5 finished with value: 0.467713583790982 and parameters: {

Best trial: 4. Best value: 0.491037:  23%|          | 7/30 [00:09<00:21,  1.06it/s]

[I 2026-04-11 22:41:40,774] Trial 6 finished with value: 0.4742961561125311 and parameters: {

Best trial: 4. Best value: 0.491037:  27%|          | 8/30 [00:12<00:29,  1.32s/it]

[I 2026-04-11 22:41:42,913] Trial 7 finished with value: 0.4754676551581094 and parameters: {

Best trial: 4. Best value: 0.491037:  30%|          | 9/30 [00:15<00:39,  1.86s/it]

[I 2026-04-11 22:41:45,955] Trial 8 finished with value: 0.4167467387764828 and parameters: {

Best trial: 4. Best value: 0.491037:  33%|          | 10/30 [00:15<00:27,  1.37s/it]

[I 2026-04-11 22:41:46,233] Trial 9 finished with value: 0.4800166350535603 and parameters: {

Best trial: 4. Best value: 0.491037:  37%|          | 11/30 [00:16<00:23,  1.24s/it]

[I 2026-04-11 22:41:47,173] Trial 10 finished with value: 0.4690773071507495 and parameters:

Best trial: 4. Best value: 0.491037:  40%|          | 12/30 [00:16<00:17,  1.02it/s]
```

```
[I 2026-04-11 22:41:47,567] Trial 11 finished with value: 0.48751448466078623 and parameters

Best trial: 4. Best value: 0.491037:  43%|        | 13/30 [00:17<00:13,  1.27it/s]

[I 2026-04-11 22:41:47,902] Trial 12 finished with value: 0.47092994153643525 and parameters

Best trial: 4. Best value: 0.491037:  47%|        | 14/30 [00:17<00:11,  1.38it/s]

[I 2026-04-11 22:41:48,491] Trial 13 finished with value: 0.4726364197021058 and parameters:

Best trial: 4. Best value: 0.491037:  50%|        | 15/30 [00:18<00:09,  1.54it/s]

[I 2026-04-11 22:41:48,962] Trial 14 finished with value: 0.47848765498245815 and parameters

Best trial: 4. Best value: 0.491037:  53%|        | 16/30 [00:19<00:13,  1.07it/s]

[I 2026-04-11 22:41:50,555] Trial 15 finished with value: 0.4671461697397321 and parameters:
[I 2026-04-11 22:41:50,649] Trial 16 finished with value: 0.42217565866907925 and parameters

Best trial: 4. Best value: 0.491037:  60%|        | 18/30 [00:20<00:07,  1.53it/s]

[I 2026-04-11 22:41:51,212] Trial 17 finished with value: 0.4785419672703098 and parameters:

Best trial: 4. Best value: 0.491037:  63%|        | 19/30 [00:22<00:12,  1.10s/it]

[I 2026-04-11 22:41:53,655] Trial 18 finished with value: 0.45097084898400547 and parameters
1, 'learning_rate': 0.031401952750335485, 'n_estimators': 537, 'min_child_samples': 131}. Bes

Best trial: 4. Best value: 0.491037:  67%|        | 20/30 [00:24<00:12,  1.29s/it]

[I 2026-04-11 22:41:55,469] Trial 19 finished with value: 0.4152543575660367 and parameters:

Best trial: 4. Best value: 0.491037:  70%|        | 21/30 [00:25<00:10,  1.12s/it]

[I 2026-04-11 22:41:56,135] Trial 20 finished with value: 0.4603633825130714 and parameters:

Best trial: 4. Best value: 0.491037:  73%|        | 22/30 [00:25<00:07,  1.08it/s]
```

```
[I 2026-04-11 22:41:56,557] Trial 21 finished with value: 0.4897292129922002 and parameters:

Best trial: 22. Best value: 0.494625:  77%|      | 23/30 [00:26<00:05,  1.32it/s]

[I 2026-04-11 22:41:56,903] Trial 22 finished with value: 0.4946246488160022 and parameters:

Best trial: 22. Best value: 0.494625:  80%|      | 24/30 [00:26<00:03,  1.64it/s]

[I 2026-04-11 22:41:57,151] Trial 23 finished with value: 0.49025001711560684 and parameters

Best trial: 22. Best value: 0.494625:  87%|      | 26/30 [00:26<00:01,  2.50it/s]

[I 2026-04-11 22:41:57,420] Trial 24 finished with value: 0.47520177760012705 and parameters
[I 2026-04-11 22:41:57,557] Trial 25 finished with value: 0.4464499112470423 and parameters:

Best trial: 22. Best value: 0.494625:  90%|      | 27/30 [00:27<00:01,  2.57it/s]

[I 2026-04-11 22:41:57,920] Trial 26 finished with value: 0.4910729951274079 and parameters:

Best trial: 22. Best value: 0.494625:  97%|      | 29/30 [00:27<00:00,  3.13it/s]

[I 2026-04-11 22:41:58,265] Trial 27 finished with value: 0.4839727137582376 and parameters:
[I 2026-04-11 22:41:58,451] Trial 28 finished with value: 0.45379099631861963 and parameters

Best trial: 22. Best value: 0.494625: 100%|      | 30/30 [00:27<00:00,  1.08it/s]

[I 2026-04-11 22:41:58,586] Trial 29 finished with value: 0.45150312295617123 and parameters
Best AUC-PR: 0.4946246488160022
Best parameters: {'num_leaves': 96, 'max_depth': 4, 'learning_rate': 0.04993468677168869, 'n
```

```python
model = lgb.LGBMClassifier(
    objective="binary",
    scale_pos_weight=scale_pos_weight,
    random_state=RANDOM_STATE,
    **best_params
)

model.fit(X_train, y_train)
```

boosting_type	'gbdt'
num_leaves	96
max_depth	4
learning_rate	0.04993468677168869
n_estimators	240
subsample_for_bin	200000
objective	'binary'
class_weight	None
min_split_gain	0.0
min_child_weight	0.001
min_child_samples	130
subsample	1.0
subsample_freq	0
colsample_bytree	1.0
reg_alpha	0.0
reg_lambda	0.0
random_state	705
n_jobs	None
importance_type	'split'
scale_pos_weight	5.802385008517888

6.2. Probability Calibration

```
# raw probabilities on validation
val_probs_raw = model.predict_proba(X_validation)[:, 1]

# isotonic regression calibration
calibrator = IsotonicRegression(out_of_bounds="clip")
calibrator.fit(val_probs_raw, y_validation)

# calibrated probabilities
val_probs_calibrated = calibrator.transform(val_probs_raw)
```

```
c3_validation_metrics = evaluate_from_probs(y_validation, val_probs_calibrated)

print(f"Cluster {cluster_id} model (validation, calibrated):")
print(c3_validation_metrics)
```

```
Cluster 3 model (validation, calibrated):
{'AUC_PR': 0.48541909654074455, 'F1': 0.4430379746835443, 'Precision': 0.6, 'Recall': 0.3511
```

```
# save the model and calibrator for cluster 3
model_dir = Path("models")
model_name = f"c{cluster_id}"

# save model
joblib.dump(model, model_dir / f"{model_name}_model.joblib")

# save calibrator
joblib.dump(calibrator, model_dir / f"{model_name}_calibrator.joblib")

# save metadata
metadata = {
    "best_params": best_params,
    "scale_pos_weight": scale_pos_weight,
    "features": list(X_train.columns),
    "random_state": RANDOM_STATE,
}

joblib.dump(metadata, model_dir / f"{model_name}_metadata.joblib")
```

```
['models/c3_metadata.joblib']
```

7. Model Comparison on the Validation Set

This section provides a unified, validation-based comparison between the calibrated global model and the calibrated cluster-specific models. By evaluating all models on the same validation data, we assess whether segmentation improves predictive performance before any test data is used.

First we load models and validation data

```
# directory with saved models
MODEL_DIR = Path("models")

# load validation data (full, with cluster labels)
X_val_full, y_val = load_dataset("02", "validation")

cluster_col = "cluster"
# feature matrix used for prediction
X_val = X_val_full.drop(columns=[cluster_col])
X_val = X_val_full.drop(columns=[cluster_col])
```

```
print(X_val.shape)
X_val.head(3)
```

```
validation
X shape: (7234, 60)
y shape: (7234,)

X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

```
(7234, 59)
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

```
# load global model and calibrator
global_model = joblib.load(MODEL_DIR / "global_model.joblib")
global_calibrator = joblib.load(MODEL_DIR / "global_calibrator.joblib")
global_metadata = joblib.load(MODEL_DIR / "global_metadata.joblib")

# ensure correct feature order
X_val_global = X_val[global_metadata["features"]]

# calibrated probabilities
global_probs_raw = global_model.predict_proba(X_val_global)[:, 1]
global_probs_cal = global_calibrator.transform(global_probs_raw)

# overall global performance
rows = []

rows.append(
```

```
    {"Model": "Global", "Scope": "All", **evaluate_from_probs(y_val, global_probs_cal)}
)
```

We do the cluster-level comparison: **Global vs Segment Models**

```
for cluster_id in sorted(X_val_full[cluster_col].unique()):
    idx = X_val_full[cluster_col] == cluster_id

    # ----- Global model on this cluster -----
    rows.append(
        {
            "Model": "Global",
            "Scope": f"Cluster {cluster_id}",
            **evaluate_from_probs(y_val.loc[idx], global_probs_cal[idx]),
        }
    )

    # ----- Cluster-specific model -----
    model = joblib.load(MODEL_DIR / f"c{cluster_id}_model.joblib")
    calibrator = joblib.load(MODEL_DIR / f"c{cluster_id}_calibrator.joblib")
    metadata = joblib.load(MODEL_DIR / f"c{cluster_id}_metadata.joblib")

    X_val_cluster = X_val.loc[idx, metadata["features"]]

    probs_raw = model.predict_proba(X_val_cluster)[:, 1]
    probs_cal = calibrator.transform(probs_raw)

    rows.append(
        {
            "Model": "Cluster-specific",
            "Scope": f"Cluster {cluster_id}",
            **evaluate_from_probs(y_val.loc[idx], probs_cal),
        }
    )
```

```
comparison_df = pd.DataFrame(rows)
comparison_df.sort_values(by=["Scope", "Model"]).reset_index(drop=True)
```

	Model	Scope	AUC_PR	F1	Precision	Recall
0	Global	All	0.446942	0.316547	0.661654	0.208038

	Model	Scope	AUC_PR	F1	Precision	Recall
1	Cluster-specific	Cluster 0	0.545604	0.446043	0.720930	0.322917
2	Global	Cluster 0	0.573545	0.444444	0.769231	0.312500
3	Cluster-specific	Cluster 1	0.278645	0.199005	0.666667	0.116959
4	Global	Cluster 1	0.292097	0.177340	0.562500	0.105263
5	Cluster-specific	Cluster 2	0.348770	0.266094	0.632653	0.168478
6	Global	Cluster 2	0.373361	0.260504	0.574074	0.168478
7	Cluster-specific	Cluster 3	0.485419	0.443038	0.600000	0.351171
8	Global	Cluster 3	0.501478	0.334165	0.656863	0.224080

The global model achieves higher AUC-PR across all clusters, indicating better overall ranking and discrimination between positive and negative cases. In contrast, cluster-specific models often achieve higher F1 and recall within individual clusters, suggesting they better capture cluster-specific patterns and improve threshold-dependent performance. This highlights a trade-off: the global model excels at general separation, while cluster-specific models can improve detection in certain segments.

8. Precision–Recall Curve Analysis

```
MODEL_DIR = Path("models")

global_model = joblib.load(MODEL_DIR / "global_model.joblib")
global_calibrator = joblib.load(MODEL_DIR / "global_calibrator.joblib")
global_metadata = joblib.load(MODEL_DIR / "global_metadata.joblib")
```

```
# load validation data (full, with cluster labels)
X_val_full, y_val = load_dataset("02", "validation")

cluster_col = "cluster"
# X_val_full already has cluster labels
X_val = X_val_full.drop(columns=[cluster_col])

# Ensure correct feature order for global model
X_val_global = X_val[global_metadata["features"]]
```

```
validation
X shape: (7234, 60)
y shape: (7234,)
```

```
X head:
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

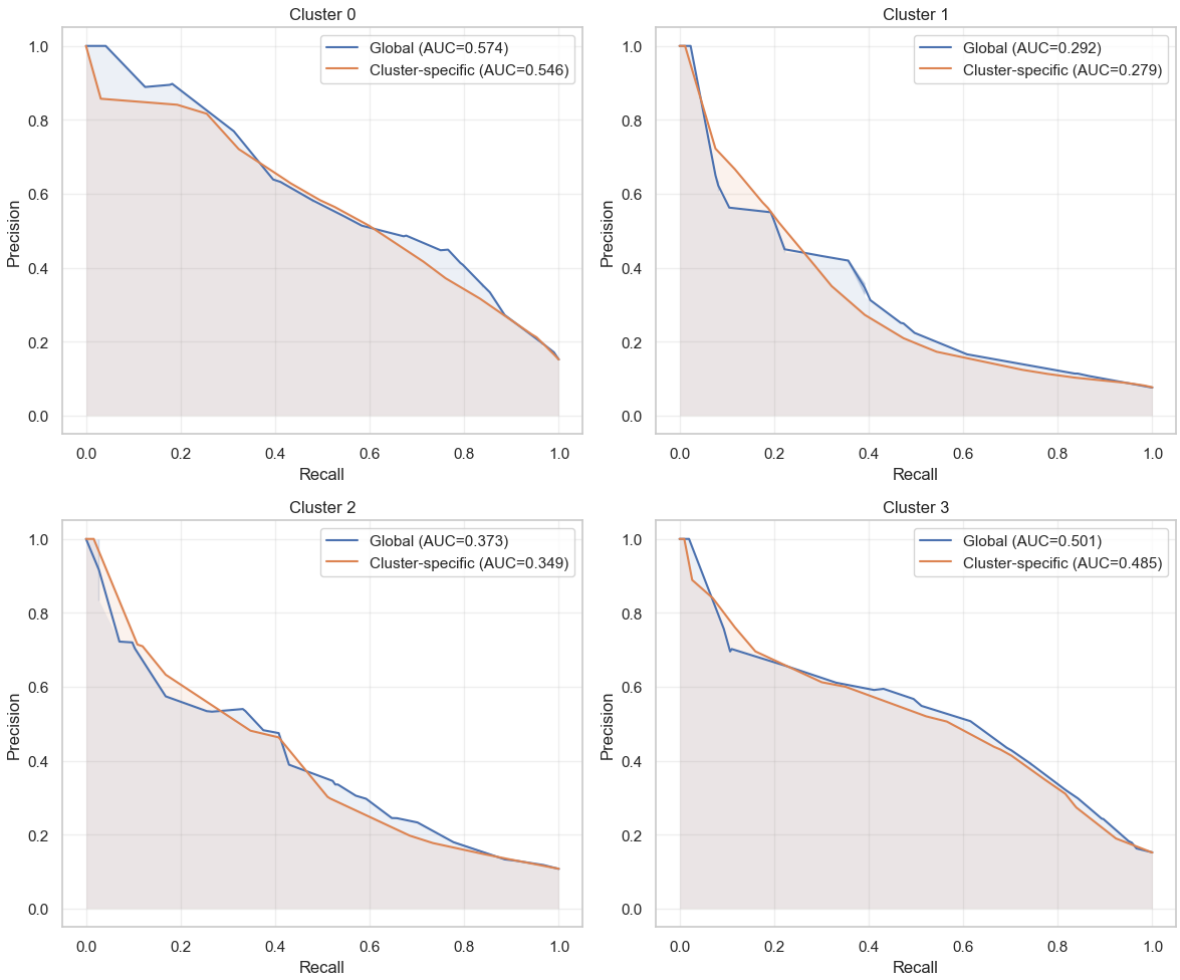
```python
# Raw probabilities
global_probs_raw = global_model.predict_proba(X_val_global)[:, 1]

# Calibrated probabilities
global_probs_cal = global_calibrator.transform(global_probs_raw)
```

```python
cluster_models = {}
cluster_calibrators = {}
cluster_metadata = {}

for cluster_id in sorted(X_val_full[cluster_col].unique()):
    cluster_models[cluster_id] = joblib.load(MODEL_DIR / f"c{cluster_id}_model.joblib")
    cluster_calibrators[cluster_id] = joblib.load(
        MODEL_DIR / f"c{cluster_id}_calibrator.joblib"
    )
    cluster_metadata[cluster_id] = joblib.load(
        MODEL_DIR / f"c{cluster_id}_metadata.joblib"
    )
```

```python
plot_cluster_pr_curves(
    global_probs_cal=global_probs_cal,
    y_val=y_val,
    X_val=X_val,
    cluster_models=cluster_models,
    cluster_calibrators=cluster_calibrators,
    cluster_metadata=cluster_metadata,
    X_val_full=X_val_full,
    cluster_col="cluster",
)
```



The precision–recall curves highlight how model performance varies across customer segments. For both the global and cluster-specific models, their curves overlap closely and none of the models dominates greatly over the other.

9. Business Cost Framework

To translate model predictions into actionable decisions, we define a simple cost-based framework reflecting the trade-offs in telemarketing campaigns.

The bank must decide which customers to contact under limited resources. Each contact incurs a cost (e.g., agent time, operational expenses), while successful subscriptions generate revenue. Therefore, the objective is to maximize the net benefit of the campaign rather than purely predictive performance.

We define: - `C_call`: cost of contacting a customer - `B_sub`: benefit from a successful subscription

This leads to the following trade-offs: - Contacting likely subscribers → generates profit - Contacting unlikely subscribers → wastes resources - Not contacting likely subscribers → missed opportunity

In expectation, a customer should be contacted only if:

$$\text{expected benefit} > \text{cost of contact}$$

This condition defines the minimum probability threshold required for a contact to be profitable.

That is, when:

$$\text{predicted probability x B_sub} > \text{C_call}$$

This framework connects model outputs (predicted probabilities) with business decisions and highlights the importance of well-calibrated probabilities, since inaccurate probabilities would lead to systematically suboptimal contact decisions.

Interpretation of Threshold-Tuned Results

When evaluating threshold tuning results, performance metrics reflect different aspects of this cost-benefit trade-off:

Metric	Interpretation
Recall	Captures more subscribers (reduces missed opportunities)
Precision	Reduces wasted calls (improves call efficiency)
F1	Balances both recall and precision objectives
PR-AUC	Reflects overall ranking quality, independent of threshold

Practical decision rule

- If campaign budget is constrained, prioritize higher precision at acceptable recall
- If growth targets dominate, prioritize higher recall subject to a minimum precision
- The selected threshold should align with operational constraints (agent capacity, call cost, and expected conversion value)

10. Threshold Selection via Expected Cost

Threshold selection is performed exclusively on the validation set to avoid overfitting. The selected threshold will be applied unchanged to the test set in the final evaluation.

```
MODEL_DIR = Path("models")
global_model = joblib.load(MODEL_DIR / "global_model.joblib")
global_calibrator = joblib.load(MODEL_DIR / "global_calibrator.joblib")
global_metadata = joblib.load(MODEL_DIR / "global_metadata.joblib")
```

```
# load validation data (full, with cluster labels)
X_val_full, y_val = load_dataset("02", "validation")

cluster_col = "cluster"
# X_val_full already has cluster labels
X_val = X_val_full.drop(columns=[cluster_col])

# Ensure correct feature order for global model
X_val_global = X_val[global_metadata["features"]]
```

```
validation
X shape: (7234, 60)
y shape: (7234,)

X head:
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__job_
0	0.0	0.0	0.0	0.0	1.0
1	0.0	1.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0

```
# Raw probabilities
global_probs_raw = global_model.predict_proba(X_val_global)[:, 1]

# Calibrated probabilities
global_probs_cal = global_calibrator.transform(global_probs_raw)
```

```
cluster_models = {}
cluster_calibrators = {}
cluster_metadata = {}

for cluster_id in sorted(X_val_full[cluster_col].unique()):
    cluster_models[cluster_id] = joblib.load(MODEL_DIR / f"c{cluster_id}_model.joblib")
    cluster_calibrators[cluster_id] = joblib.load(
        MODEL_DIR / f"c{cluster_id}_calibrator.joblib"
    )
    cluster_metadata[cluster_id] = joblib.load(
        MODEL_DIR / f"c{cluster_id}_metadata.joblib"
    )
```

The threshold that maximizes expected profit is selected, aligning model decisions with the business objective of maximizing campaign net benefit rather than purely predictive performance.

1. Define business parameters

We begin by defining the cost-benefit parameters used in the decision framework. These values represent the relative cost of contacting a customer and the benefit of a successful subscription. We also compute the theoretical break-even probability threshold, which provides a baseline reference for decision-making.

```
# Business parameters
C_call = 1.0  # cost per call
B_sub = 10.0  # benefit per subscription

# Theoretical break-even threshold
break_even_p = C_call / B_sub
# Only contact customers with  break_even_p% probability of subscribing
print(f"Break-even probability: {break_even_p:.3f}")
```

```
Break-even probability: 0.100
```

The cost and benefit parameters are normalized to reflect the relative magnitude of contacting a customer versus obtaining a successful subscription. We assume that the benefit of a subscription is approximately x times the cost of a call.

Expected profit is computed using predicted probabilities as estimates of conversion likelihood, allowing us to approximate the expected value of contacting each customer.

2. Global threshold selection

We first determine the optimal threshold for the global model using the full validation set. For a range of candidate thresholds, we convert predicted probabilities into binary decisions and evaluate both classification metrics and expected profit. The optimal threshold is selected as the one that maximizes expected profit.

```
thresholds = np.linspace(0.01, 0.5, 50)

global_results = []

for t in thresholds:
    y_pred = (global_probs_cal >= t).astype(int)

    precision = precision_score(y_val, y_pred, zero_division=0)
    recall = recall_score(y_val, y_pred)
    f1 = f1_score(y_val, y_pred)

    contacted = global_probs_cal >= t
    expected_profit = np.sum(global_probs_cal[contacted] * B_sub - C_call)

    n_contacted = contacted.sum()
    pct_contacted = n_contacted / len(y_val)

    global_results.append(
        {
            "threshold": t,
            "precision": precision,
            "recall": recall,
            "f1": f1,
            "expected_profit": expected_profit,
            "n_contacted": n_contacted,
            "pct_contacted": pct_contacted,
        }
    )

global_results = pd.DataFrame(global_results)
global_results["pct_contacted"] = global_results["pct_contacted"] * 100

# Best threshold
best_global_row = global_results.loc[global_results["expected_profit"].idxmax()]
best_threshold_global = best_global_row["threshold"]
```

```
print("Best GLOBAL threshold:", round(best_threshold_global, 4))
display(best_global_row)
```

```
Best GLOBAL threshold: 0.07

threshold             0.070000
precision             0.277289
recall                0.744681
f1                    0.404105
expected_profit    4028.000000
n_contacted        2272.000000
pct_contacted        31.407244
Name: 6, dtype: float64
```

3. Cluster-specific threshold selection

We now repeat the threshold optimization procedure for each cluster-specific model. For each segment, we restrict the validation data to observations belonging to that cluster and evaluate thresholds using the corresponding model and calibrated probabilities. This allows the decision rule to adapt to differences in behavior across customer segments.

```
cluster_thresholds = {}
cluster_results = {}

for cluster_id in sorted(X_val_full[cluster_col].unique()):

    print(f"\nProcessing cluster {cluster_id}")

    # Filter validation data
    mask = X_val_full[cluster_col] == cluster_id

    X_val_k = X_val_full[mask].drop(columns=[cluster_col])
    y_val_k = y_val[mask]

    # Ensure correct feature order
    X_val_k = X_val_k[cluster_metadata[cluster_id]["features"]]

    # Get model + calibrator
    model_k = cluster_models[cluster_id]
    calibrator_k = cluster_calibrators[cluster_id]
```

```python
# Probabilities
probs_raw_k = model_k.predict_proba(X_val_k)[:, 1]
probs_cal_k = calibrator_k.transform(probs_raw_k)

# Threshold search
results_k = []

for t in thresholds:
    y_pred_k = (probs_cal_k >= t).astype(int)

    precision = precision_score(y_val_k, y_pred_k, zero_division=0)
    recall = recall_score(y_val_k, y_pred_k)
    f1 = f1_score(y_val_k, y_pred_k)

    contacted = probs_cal_k >= t
    expected_profit = np.sum(probs_cal_k[contacted] * B_sub - C_call)

    n_contacted = contacted.sum()
    pct_contacted = n_contacted / len(y_val_k)

    results_k.append(
        {
            "threshold": t,
            "precision": precision,
            "recall": recall,
            "f1": f1,
            "expected_profit": expected_profit,
            "n_contacted": n_contacted,
            "pct_contacted": pct_contacted,
        }
    )

results_k = pd.DataFrame(results_k)
results_k["pct_contacted"] = results_k["pct_contacted"] * 100

# Best threshold for this cluster
best_row_k = results_k.loc[results_k["expected_profit"].idxmax()]
best_t_k = best_row_k["threshold"]

cluster_thresholds[cluster_id] = best_t_k
cluster_results[cluster_id] = results_k
```

```
    print(f"Best threshold for cluster {cluster_id}: {best_t_k:.4f}")
    display(best_row_k)
```

```
Processing cluster 0
Best threshold for cluster 0: 0.0700

threshold             0.070000
precision             0.316206
recall                0.833333
f1                    0.458453
expected_profit    1094.000000
n_contacted         506.000000
pct_contacted        39.811172
Name: 6, dtype: float64


Processing cluster 1
Best threshold for cluster 1: 0.1000

threshold            0.100000
precision            0.272358
recall               0.391813
f1                   0.321343
expected_profit    424.000000
n_contacted        246.000000
pct_contacted       10.856134
Name: 9, dtype: float64


Processing cluster 2
Best threshold for cluster 2: 0.0800

threshold            0.080000
precision            0.239407
recall               0.614130
f1                   0.344512
expected_profit    658.000000
n_contacted        472.000000
pct_contacted       27.441860
Name: 7, dtype: float64
```

```
Processing cluster 3
Best threshold for cluster 3: 0.0600

threshold             0.060000
precision             0.310433
recall                0.816054
f1                    0.449770
expected_profit    1654.000000
n_contacted         786.000000
pct_contacted        39.757208
Name: 5, dtype: float64
```

4. Final thresholds

Finally, we summarize the selected thresholds for both the global model and the cluster-specific models. These thresholds define the decision rules that will be applied to the test set in the final evaluation.

```python
threshold_table = pd.DataFrame(
    {
        "Model": ["Global"] + [f"Cluster {k}" for k in cluster_thresholds.keys()],
        "Threshold": [best_threshold_global] + list(cluster_thresholds.values()),
        "% Contacted": [best_global_row["pct_contacted"]]
        + [
            cluster_results[k]
            .loc[
                cluster_results[k]["threshold"] == cluster_thresholds[k],
                "pct_contacted",
            ]
            .values[0]
            for k in cluster_thresholds.keys()
        ],
        "Expected Profit": [best_global_row["expected_profit"]]
        + [
            cluster_results[k]
            .loc[
                cluster_results[k]["threshold"] == cluster_thresholds[k],
                "expected_profit",
            ]
            .values[0]
```

```
            for k in cluster_thresholds.keys()
        ],
    }
)

threshold_table["Threshold"] = threshold_table["Threshold"].round(4)
threshold_table["% Contacted"] = threshold_table["% Contacted"].round(2)

display(threshold_table)
```

	Model	Threshold	% Contacted	Expected Profit
0	Global	0.07	31.41	4028.0
1	Cluster 0	0.07	39.81	1094.0
2	Cluster 1	0.10	10.86	424.0
3	Cluster 2	0.08	27.44	658.0
4	Cluster 3	0.06	39.76	1654.0

```
# Save thresholds for later use
path = Path("models")
joblib.dump(best_threshold_global, path / "global_threshold.joblib")
joblib.dump(cluster_thresholds, path / "cluster_thresholds.joblib")
```

```
['models/cluster_thresholds.joblib']
```

11. Summary of Validation-Stage Findings

This section summarizes the key findings from the validation stage, integrating model performance, probability calibration, and threshold-based decision analysis within the proposed business framework.

1. Model Performance and Selection

Among the candidate models evaluated, **LightGBM** demonstrated the strongest overall performance in terms of AUC-PR, indicating superior ability to rank likely subscribers in the presence of class imbalance. This justified its selection as the primary modeling approach for both global and segment-specific analyses.

2. Calibration and Probability Reliability

Probability calibration was applied to improve the reliability of predicted probabilities. This step is critical because downstream decisions—specifically whether to contact a customer—depend directly on these probabilities. By aligning predicted probabilities with observed outcomes, calibration ensures that the expected profit framework produces meaningful and interpretable results.

3. Global Decision Strategy

4. Segment-Specific Decision Strategies

5. Global vs. Segmented Strategy

6. Business Interpretation